Jiali Cheng

jiali_cheng@uml.edu | Google Scholar | Personal Website

SUMMARY

PhD studying LLM, Agent, Multi-Agent, Post-Training, RL, Trustworthy AI, AI for Science

2-time Applied Scientist Intern at Amazon Alexa AI on Web Agents / Multi-Agent Evaluation

3-year Data Scientist / Engineer building AI for Bio/Health products at a Khosla Venture-backed startup

EDUCATION

UMass Lowell, USA Ph.D. in Computer Science — 2023 - present

Northeastern University, Boston, USA M.S. in Computer Engineering — 2017 - 2019

EXPERIENCE

Amazon Alexa AI — May - Aug, 2026

Applied Science Intern   Advised by: Oleg Rokhlenko, Xingzhi Guo, Jialiang Xu

- Proposed a novel setting and benchmark & environment to evaluate long-horizon decentralized multi-agent collaboration at large scale
- Designed graph-aware harness to improve overtime (RSI); Train small models with RL to improve collaboration, outperforming GPT-5.5 and Claude Opus 4.8.

Amazon Alexa AI — May - Aug, 2025

Applied Science Intern   Advised by: Oleg Rokhlenko, Anjishnu Kumar

- Designed web agents with episodic memory, efficient action-simulation, and environment exploration to support adaptive decision-making and prevent errors. Achieved SOTA results on WebArena-Lite.

Computational Language Understanding Lab, UMass Lowell — 2023 - Present

Research Assistant   Advised by: Hadi Amiri

- Built benchmarks to evaluate LLM web agents on urban / civil domain websites and tasks
- Built benchmarks to evaluate knowledge conflicts when agent interacts with environment (tool-memory conflict, skill conflict, instruction conflict)
- Develop post-training methods (RL, DPO, GRPO, OPD) to make AI models more trustworthy and controllable (unlearning, editing, robustness, bias and fairness)
- Design mechanistic interpretability methods (SAE, circuits, natural language autoencoder NLA) to 1) edit AI models to produce less harmful responses, 2) control LLM reasoning effort

Foodome Inc., Data Scientist / Data Engineer — 2020 - 2022

· Built NLP and KG methods to mine large-scale literature, build domain-specific knowledge graphs and predict new relationships, increasing Acc by 53%+

· Managed reproducible AWS workflows for data acquisition, knowledge extraction, relation prediction, and drug repurposing pipeline (Terraform, GLUE, SageMaker)

Zitnik Lab, Harvard University 2022 - 2023
Research Assistant Advised by: Marinka Zitnik

Barabasi Lab, Network Science Institute 2018 - 2020
Research Associate Advised by: Albert-Laszlo Barabasi

Research Computing Team, NEU, Assistant 2018 - 2020

- Manage High-Performance Computing cluster; Monitor, install hardware, software; Teach hands-on tutorials
- Help users parallelize code in different programming languages to run faster; Max speedup 10.3x

AWARDS

2025 Outstanding Graduate Research Award for Ph.D. Students, UMass Lowell School of CS

PUBLICATIONS

Workshop & Preprint

[1] TRACER: Token ReAssignment for Concept ERasure in Generative Recommendation
Ziheng Chen, **Jiali Cheng**, Zezhong Fan, Hadi Amiri, Diyuan Wu, Gabriele Tolomei, Yang Zhang
arXiv Link

[2] WebATLAS: Agentic Task-completion with Look-ahead Action Simulation
Jiali Cheng, Anjishnu Kumar, et al. LAW @ NeurIPS 2025 Link

[3] Investigating Tool-Memory Conflict in Tool-Augmented LLMs
Jiali Cheng, Rui Pan, Hadi Amiri R2F @ ICML 2025 Link

[4] Do Students Debias Like Teachers? On the Distillability of Bias Mitigation Methods
Jiali Cheng, Chirag Agarwal, Hadi Amiri R2F @ ICML 2025 Link

[5] Linguistic Blind Spots of Large Language Models
Jiali Cheng, Hadi Amiri CMCL Workshop @ NAACL 2025 Link

[6] MU-Bench: A Multi-task Multi-modal Benchmark for Machine Unlearning
Jiali Cheng, Hadi Amiri SafeGenAI Workshop @ NeurIPS 2024 Link

[7] CURE: Circuit-Aware Unlearning for LLM-based Recommendation
Ziheng Chen, Zezhong Fan, **Jiali Cheng**, Hadi Amiri, Yunzhi Yao, Yang Zhang, Xiangguo Sun
arXiv Link

Conference

[8] Understanding Machine Unlearning Through The Lens of Mode Connectivity
Jiali Cheng, Hadi Amiri COLM 2026 Link

[9] A Mechanistic Perspective and Circuit-Guided Difficulty Metric for Unlearning
Jiali Cheng, Ziheng Chen, Chirag Agarwal, Hadi Amiri ACL 2026 Findings Link

[10] Fair Cognitive Impairment Detection Through Unlearning
William Nguyen, **Jiali Cheng**, Hadi Amiri Interspeech 2026 Link

[11] FUTURE: Flexible Unlearning for Tree Ensemble
Ziheng Chen, Jin Huang, **Jiali Cheng**, Yuchan Guo, Mengjie Wang, Lalitesh Morishetti, Kaushiki Nag, Hadi Amiri   CIKM 2025 Link

[12] FROG: Fair Removal on Graphs
Ziheng Chen, **Jiali Cheng**, Gabriele Tolomei, Sijia Liu, Hadi Amiri, Yu Wang, Kaushiki Nag, Lu Lin
CIKM 2025 Link

[13] Speech Unlearning
Jiali Cheng, Hadi Amiri   Interspeech 2025, Blue Sky Track Link

[14] Tool Unlearning for Tool Augmented LLMs
Jiali Cheng, Hadi Amiri   ICML 2025 Link

[15] EqualizeIR: Mitigating Linguistic Biases in Retrieval Models
Jiali Cheng, Hadi Amiri   NAACL 2025 (Short) Link

[16] FairFlow: Mitigating Dataset Biases through Undecided Learning for NLU
Jiali Cheng, Hadi Amiri   EMNLP 2024 (Long) Link

[17] MultiDelete for Multimodal Machine Unlearning
Jiali Cheng, Hadi Amiri   ECCV 2024 Link

[18] MedDec: A Dataset for Extracting Medical Decisions from Discharge Summaries
Mohamed Elgaar, **Jiali Cheng**, Nidhi Vakil, Hadi Amiri, Leo Anthony Celi
Findings of ACL 2024 (Long) Link

[19] CogniVoice: Multimodal and Multilingual Fusion Networks for Mild Cognitive Impairment Assessment from Spontaneous Speech
Jiali Cheng, Mohamed Elgaar, Nidhi Vakil, Hadi Amiri   Interspeech 2024 Link

[20] Exploring the Impact of Model Scaling on Parameter-Efficient Tuning
Yusheng Su, Chi-Min Chan, **Jiali Cheng**, Yujia Qin, Yankai Lin, Shengding Hu, Zonghan Yang, Ning Ding, Xingzhi Sun, Guotong Xie, Zhiyuan Liu, Maosong Sun
EMNLP 2023 (Long) Link

[21] GNNDelete: A General Strategy for Unlearning in Graph Neural Networks
Jiali Cheng, George Dasoulas, Huan He, Chirag Agarwal, Marinka Zitnik   ICLR 2023 Link

Journal

[22] Language-Specific Representation of Emotion-Concept Knowledge Causally Supports Emotion Inference
Ming Li, Yusheng Su, Hsiu-Yuan Huang, **Jiali Cheng**, Xin Hu, Xinmiao Zhang, Huadong Wang, Yujia Qin, Xiaozhi Wang, Zhiyuan Liu, Dan Zhang
iScience Volume 27, Issue 12, 111401, Cell Press Link

TEACHING & TALKS

Machine Learning   TA, UMass Lowell, Fall 2025

Graph Machine Unlearning Guest Lecture, Advanced Social Computing Course @ UMass Lowell, Spring 2023

Computing I & II Lab (C Programming) TA, UMass Lowell, 2023, 2024

Introduction to HPC Cluster with SLURM Lecturer, Northeastern University, 2018 - 2019

Introduction to Python Lecturer, Northeastern University, 2018 - 2019

SERVICE

- Committee for SemEval Task Proposals 2027

Reviewer:

- ICLR since 2024, COLM since 2026, ARR since 2024
- KDD since 2025, Interspeech since 2026
- IEEE TPAMI, WACV 2023, EMNLP 2022, NAACL 2022, IEEE Vis 2021, 2022

Mentoring:

- William Nguyen, BS @ UMass Lowell
- Carlo Reyes, BS @ UMass Lowell
- Anugrah Vaishnav, MS @ UMass Lowell
- HaEun Kim, MS @ UMass Lowell
- Tolotra Samuel Randriakotonjanahary, MS @ UMass Lowell
- Camila Camacho, BS @ UMass Lowell